Data transformations and data visualization

Overview

Data transformations using dplyr

- Overview of the main functions in dplyr
- Using dplyr to analyze movie ratings

Data visualization using ggplot

- Conceptual overview of the grammar of graphics
- If there is time:
  - Using ggplot to create data visualizations

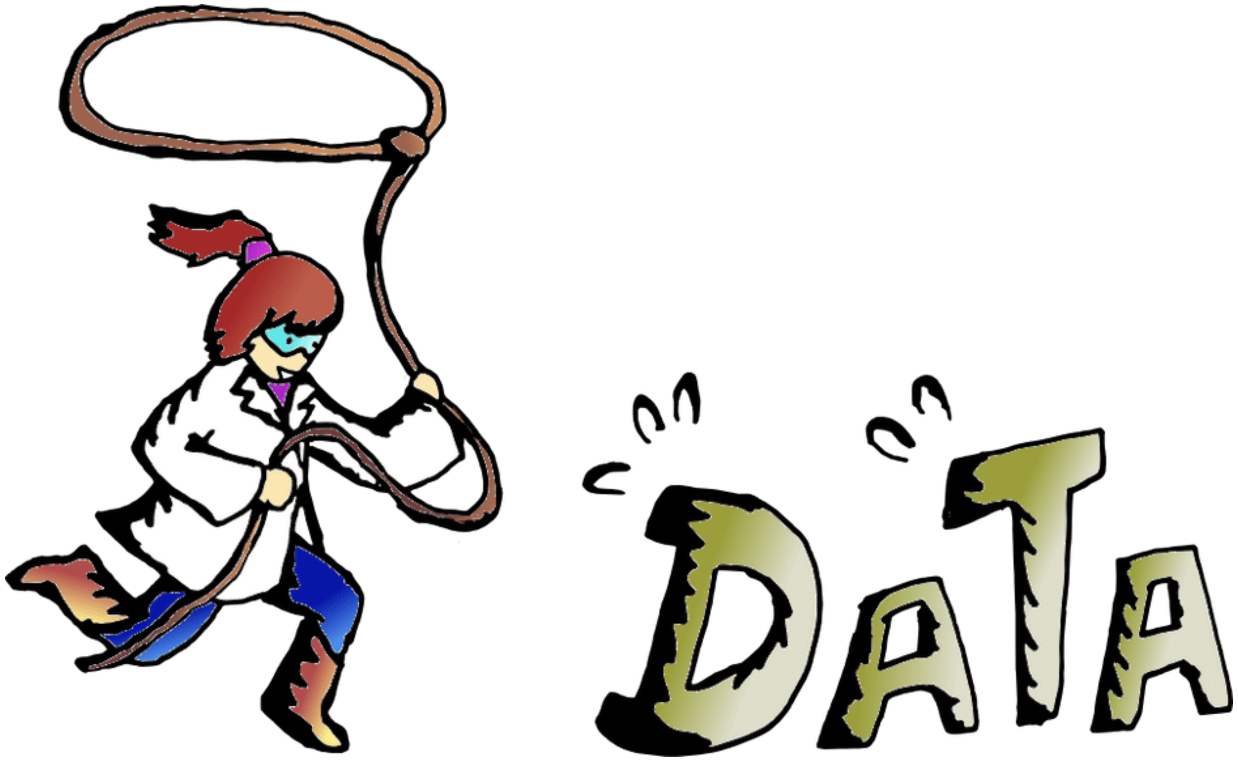
Homework 3

It is due on Gradescope by 11pm on Monday July 20st

How did homework 2 go?

Questions about anything before we start?

Data wrangling/transformation using dplyr



The ‘tidyverse’

The tidyverse is set of R packages that operate ‘tidy data’

- i.e., that operate on data frames (or tibbles)

Tidy data is data where:

- Each variable must have its own column
- Each observation must have its own row
- Each value must have its own cell

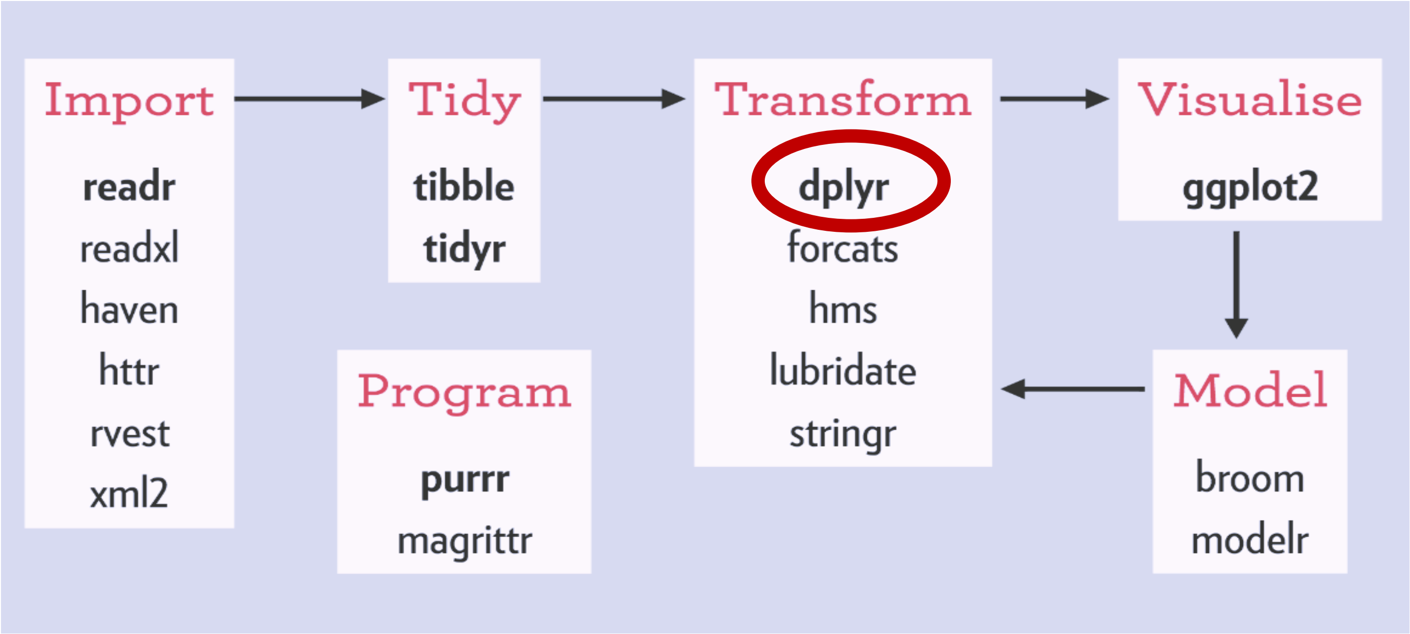
Messy data…

Messy data can be difficult to deal with

The ‘tidyverse’

The tidyverse is a set of packages share a common design philosophy

- Most written by Hadley Wickham



dplyr: A grammar for data wrangling

Grammar: a set of components that can be combined to achieve a goal

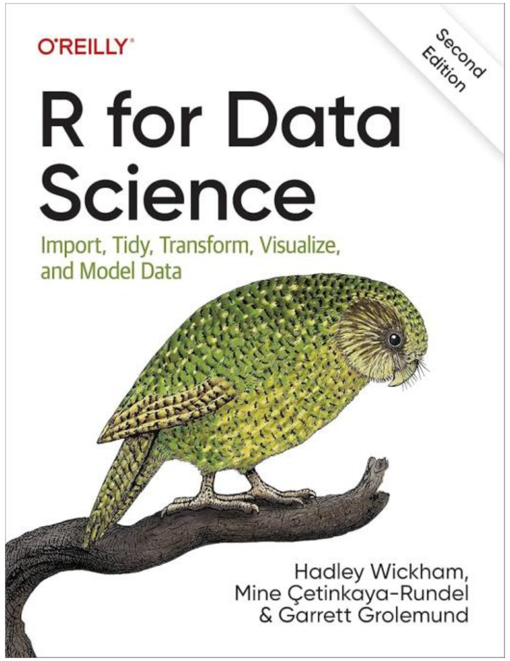
dplyr is a package that has a set of verbs that are useful for transformations data:

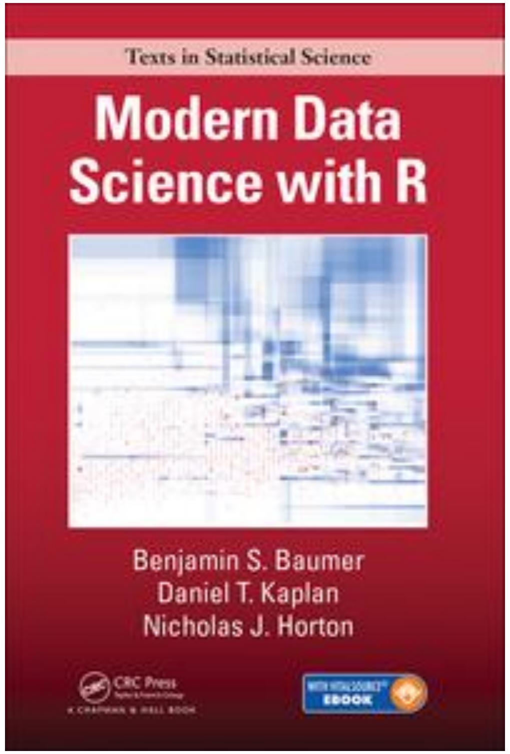
1. filter()
2. select()
3. mutate()
4. arrange()
5. group_by()
6. summarize()

All these function **take a data frame** and other arguments and **return a data frame**

```
> library(dplyr)   # load the dplyr package
```

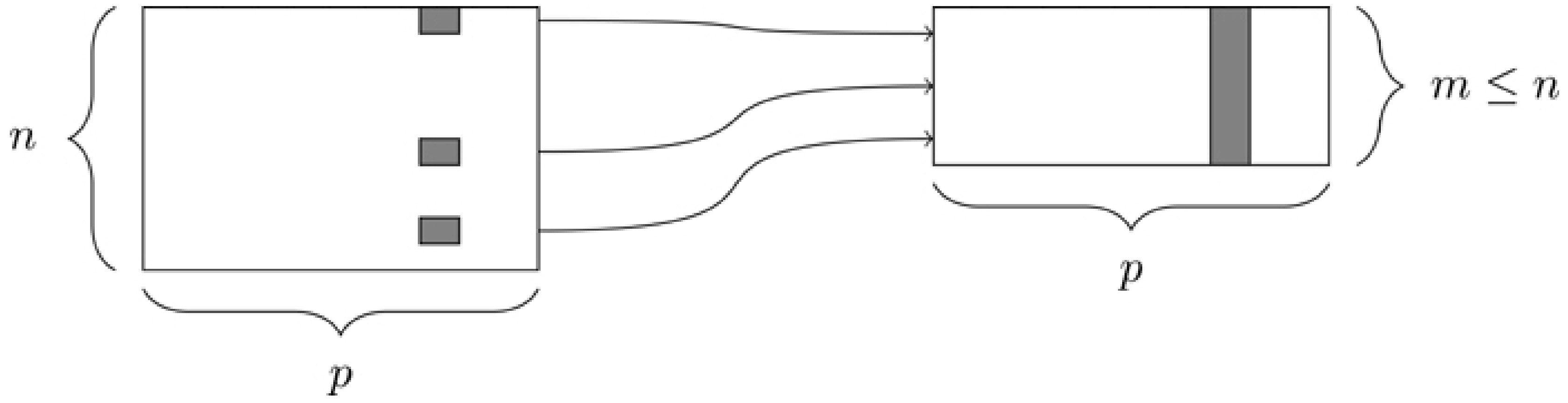
Quick overview of the dplyr functions





1. filter()

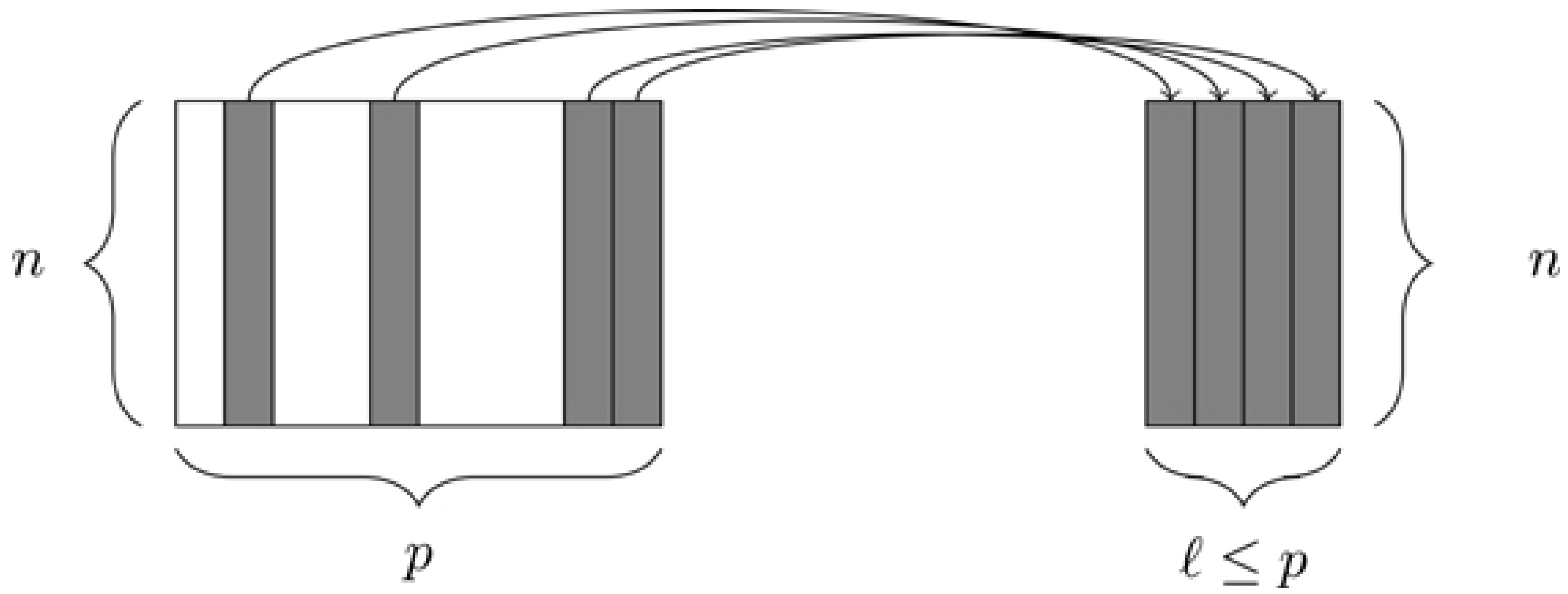
The filter() function allows you to select a subset of rows in data frame



filter(profiles, height == 77)

2. select()

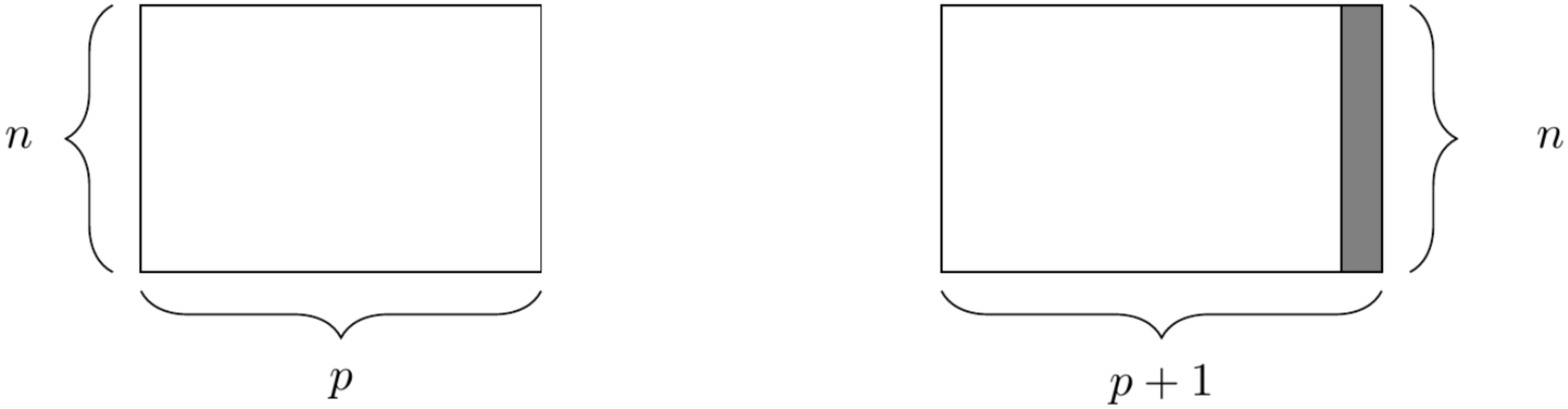
The select() function allows you to select a subset of columns



select(profiles, age, height)

3. mutate()

The mutate() function allows you to create new columns that are functions of existing columns

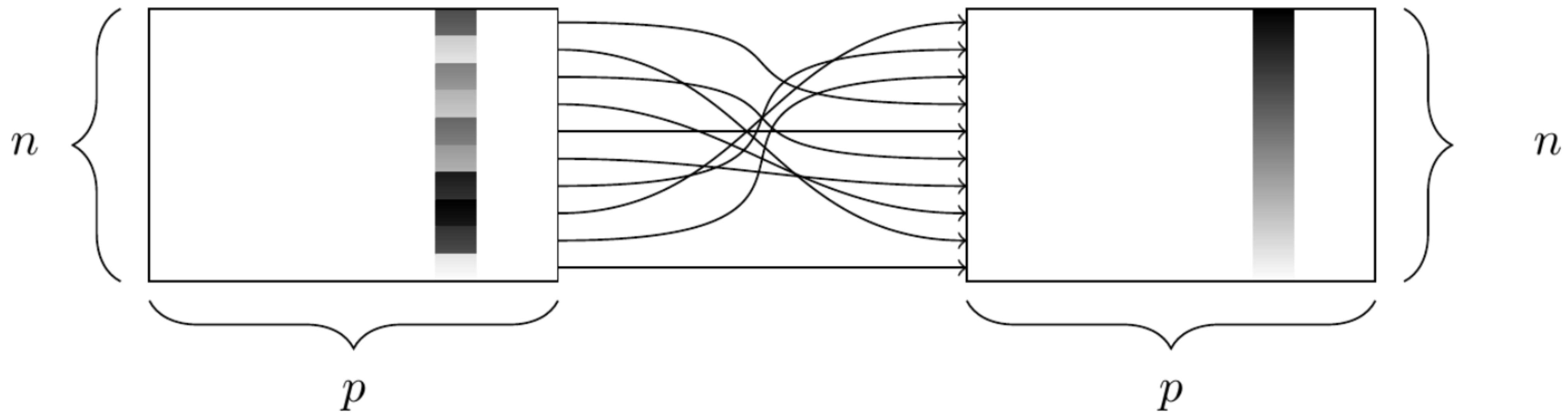


mutate(profiles, height_feet = height/12)

4. arrange()

The arrange() function arranges the rows based values in a column

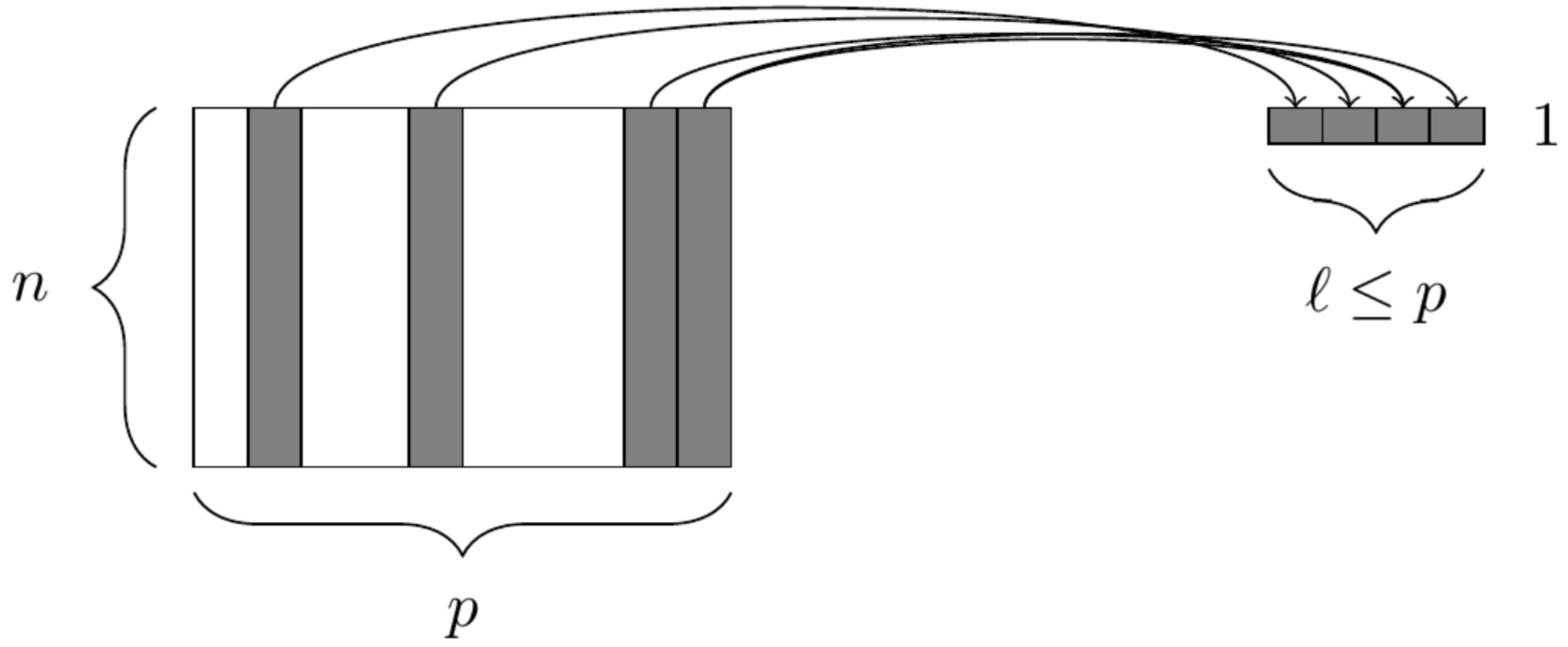
- arrange(desc()) arranges from largest to smallest



arrange(profiles, desc(height))

5. summarize()

The summarize() function reduces values in many rows into single values

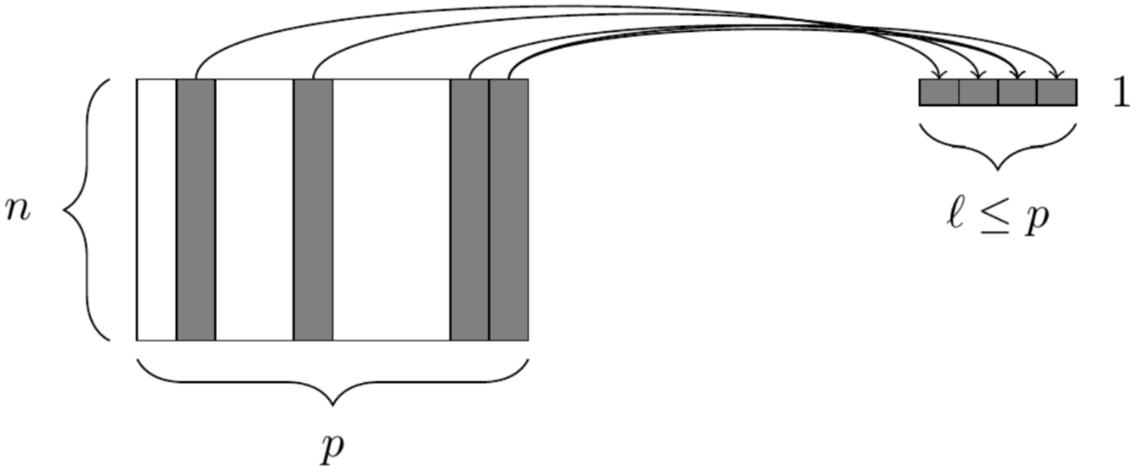


summarize(mean_age = mean(age))

6. The group_by() function

The group_by() function groups variables for future operations

- It works in conjunction with summarize() and mutate()
- It is used to do split, apply, combine



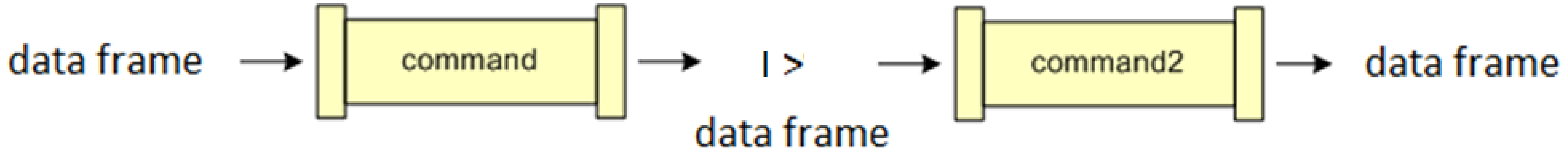
Input Data

x	y
a	2
a	4
b	0
b	5
c	5
c	10

group_by(profiles, drinks)

The pipe operator

The pipe operator |> allows us to chain commands together

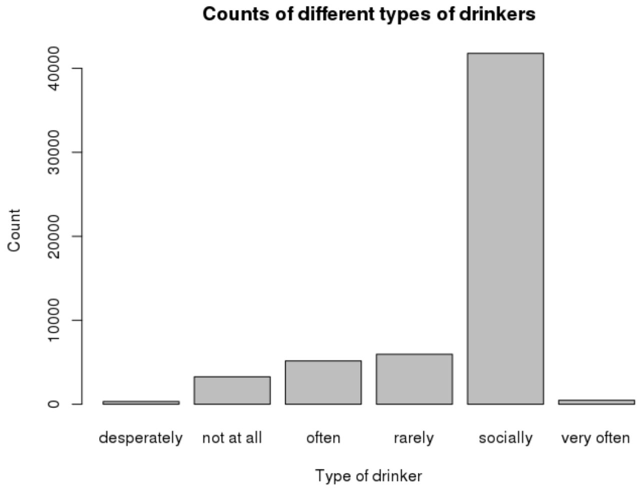


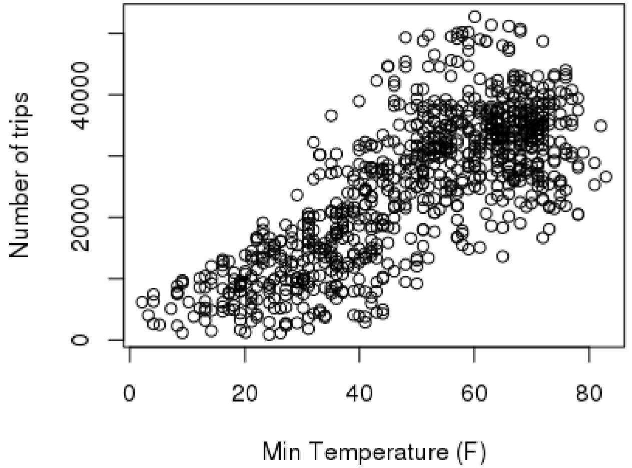

```
profiles|>
        group_by(drinks) |>
        summarize(mean_age = mean(age))
```

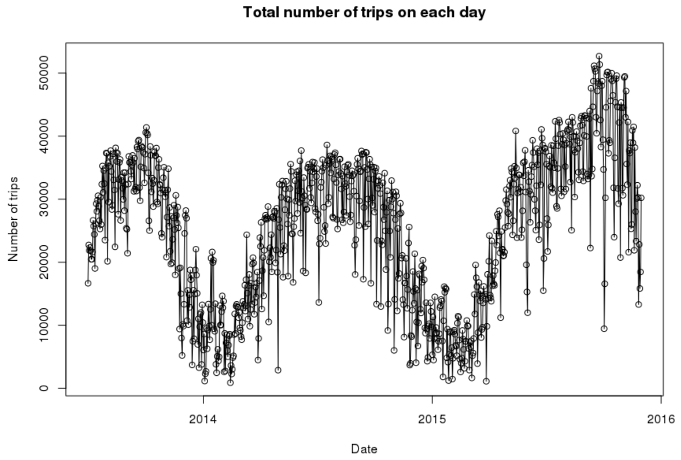
Let’s try it out!

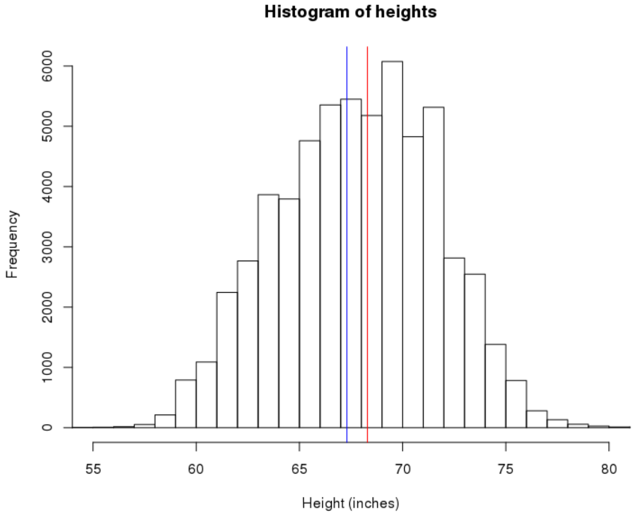
A grammar of graphics and ggplot

What are some similarities between these graphs?







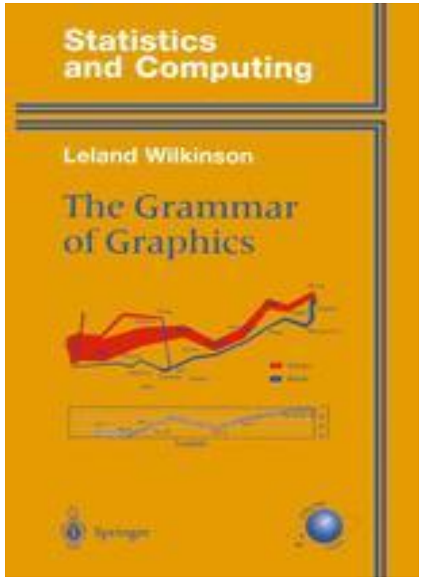


The grammar of graphics

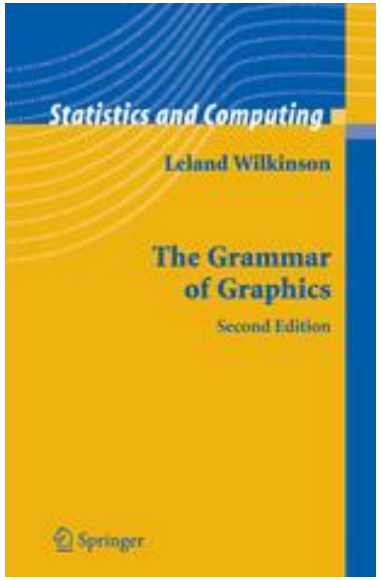
Leland Wilkinson noticed similarities between many graphs and tried to generate a ‘grammar’ that could be used to express a graph

- i.e., a list elements that can be combined together to create a graph

First edition



Second edition



Graphs are composed of…

A Frame: Coordinate system on which data is placed

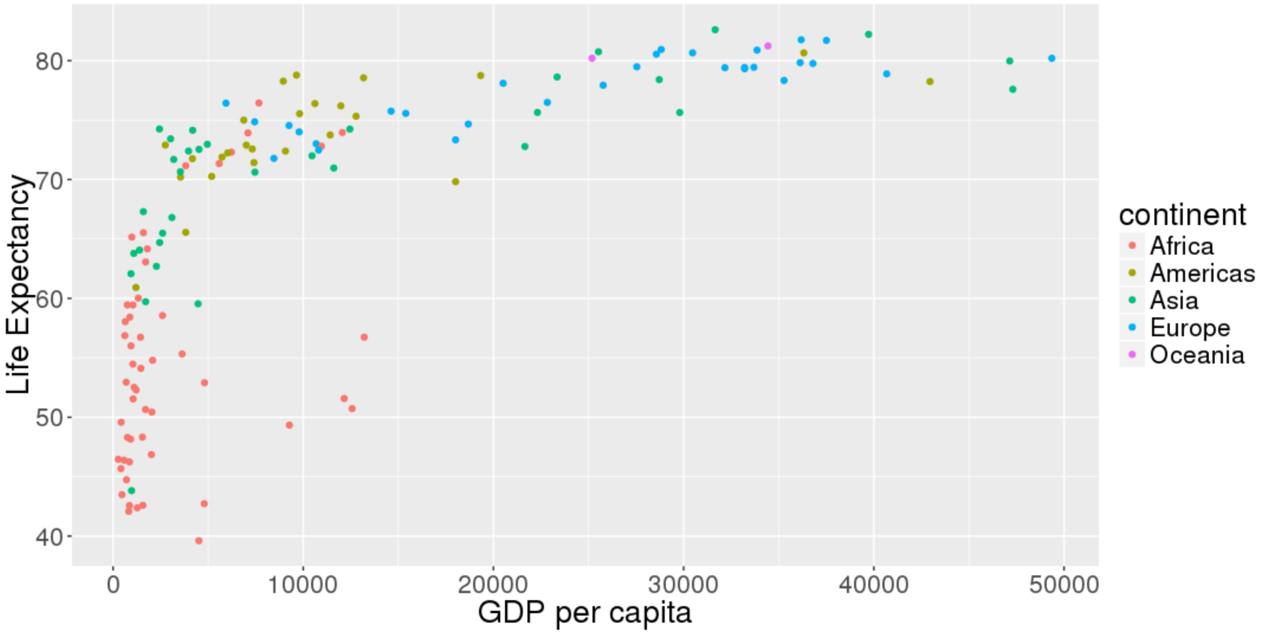
- E.g., Cartesian coordinate system, polar coordinates, etc.

Glyphs: basic graphic unit representing cases or statistics

- Contains visual properties (aesthetics) such as: position, shape, color, size, etc.
- Need to specify data values are **mapped** onto these aesthetics properties

Scales and guides: shows how to interpret axes and other properties of the glyphs

- i.e., tells us how the data values are mapped into glyph properties



Plots can also contain…

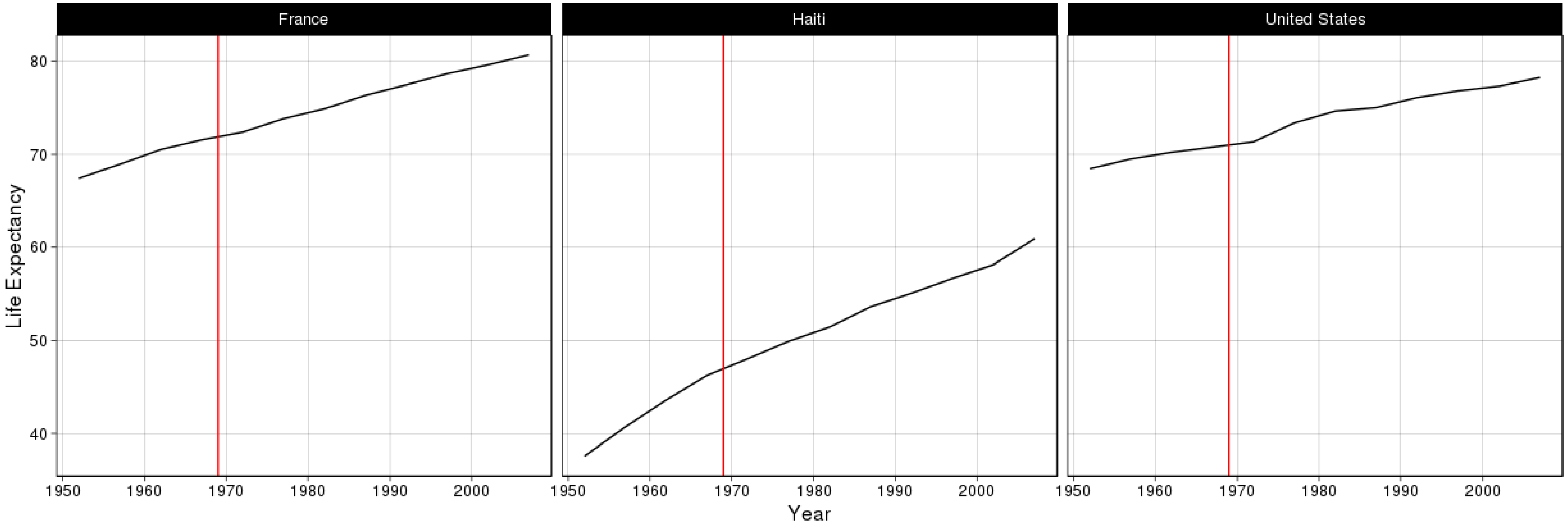
Facets: allows for multiple side-by-side graphs based on a categorical variable

- Makes it easier to compare different conditions

Layers: allows for more than one types of data to be mapped onto the same figure

Theme: contains finer points of display

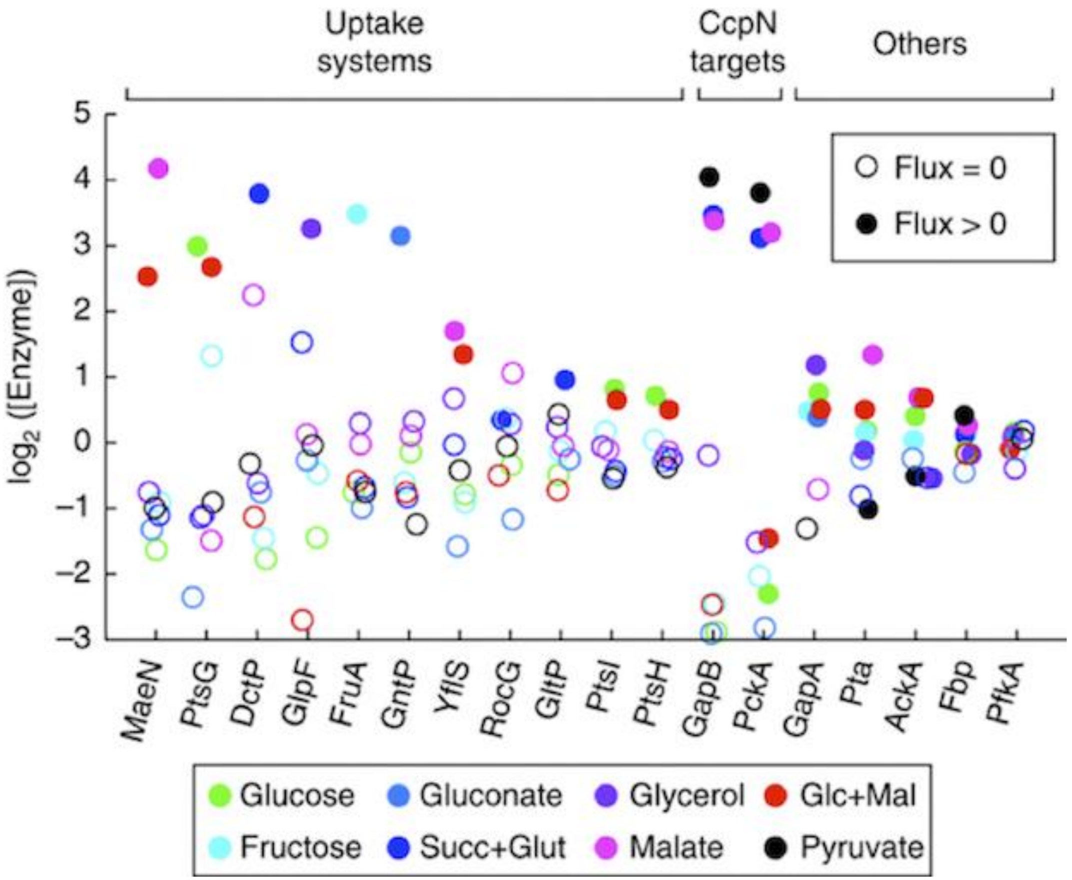
- E.g., font size, background color, etc.



Aesthetic mappings

Data Frame

Log_Enzyme	Gene	Target	Flux	Molecule
4.5	MaeN	Uptake	Positive	Glucose
-0.2	PtsG	Uptake	Zero	Glucose
3.8	PckA	CcpN	Positive	Pyruvate
-0.1	MaeN	Uptake	Zero	Gluconate
3.2	GntP	Uptake	Positive	Glycerol



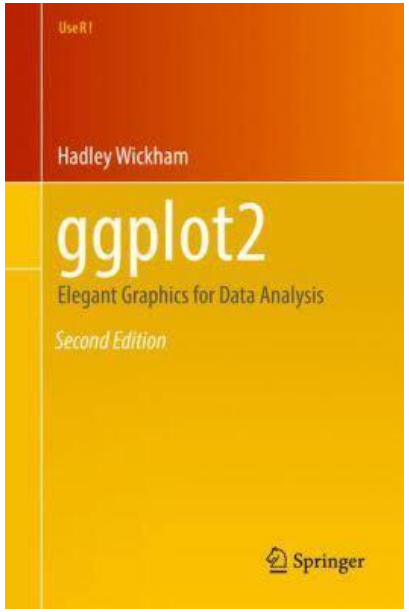
Q: What are the mappings between each variable and visual attribute?

ggplot

ggplot2 is an R package that implements the grammar of graphics

- It builds up graphics by starting with a frame, adding glyphs, etc.

```
# load the ggplot2 library
> library('ggplot2')
```



Get the book on GitHub

Example data: mtcars

ROAD TEST By Jim Brokaw

THE LUXURY CARS

Imperial Palace
Fortress Fleetwood
Castle Continental

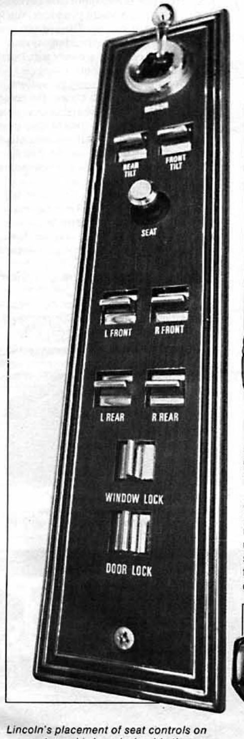
Crippled by the fuel flap and sniggered at lewdly by those smug Mercedes owners, today it seems that these great mastadons are dismissed as symbols of an ancient aristocracy whose strata was marked by expanse of wheelbase, and the heavenly quantity of gross cubic mass able to be shouldered by four beleagured tires. As the sole surviving heirs of princely Packards, dynosaurean Duesenbergs and the Brobdingnagian Bugattis, these marvels of grand proportion should be headed for the Smithsonian Institute by way of the mucky La Brea tar pits.

But are they?

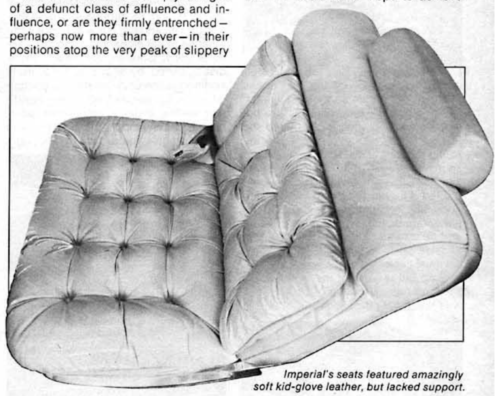
Are these behemoths simply vestiges of a defunct class of affluence and influence, or are they firmly entrenched—perhaps now more than ever—in their positions atop the very peak of slippery Mount Status, whose crags we scale daily, whether we wish to acknowledge that fact or not?

Ah, but welcome to the current state of American affairs: beef prices manipulated by withholding the animals from market; endless rounds of strikes by unions whose members are engaged in serving the public; gasoline supplies that rise and fall in mysterious coincidence with rising prices; dry rot, manipulation and partisan hatcheteering in government. They leave us little to believe in, and less to trust.

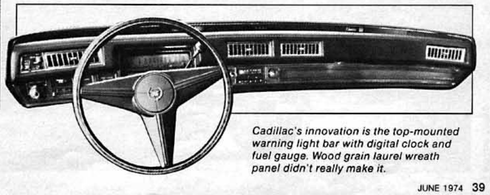
The very qualities that appear to condemn these sail-less luxury liners will very likely ensure their perpetuation. In days past, the block-long Cadillac and shiny Lincoln with paint jobs three feet deep flaunted a socio-economic position we could never hope to achieve. They did, however, constantly remind us that such positions, such wealth, such power, did, in fact, exist. The gliding specter of the shiny Imperial eagle stirred within a few heretical souls the bold idea that if such positions of power and wealth existed, there must be some means of attainment. More than a few of the haughty, distant drivers of the velvet tanks clawed their way up from the very pavement they now whisper over to



Lincoln's placement of seat controls on arm rest panel is less desireable than lower side-of seat location of Cad and Imperial.



Imperial's seats featured amazingly soft kid-glove leather, but lacked support.



Cadillac's innovation is the top-mounted warning light bar with digital clock and fuel gauge. Wood grain laurel wreath panel didn't really make it.

PERFORMANCE	CADILLAC	LINCOLN	IMPERIAL
Acceleration 0-30 mph	4.30	3.97	4.2
0-50 mph	8.49	8.00	9.15
0-60 mph	12.00	9.50	12.1
Standing Start 1/4-mile Mph	77.05	77.65	80.28
Elapsed time	17.98	17.82	17.42
Passing speeds 40-60 mph	6.58	5.9	7.1
50-70 mph	7.00	6.8	6.8
Stopping distance From 30 mph	32′1″	31′4″	27′5″
From 60 mph	182′7″	153′10″	129′3″
Gas mileage range	10.43	10.42	14.7
Width—in.	79.8	80.0	79.7
Front Track—in.	63.5	64.3	64
Rear Track—in.	63.3	64.3	63.7
Wheelbase—in	133.0	127.0	124.0
Overall length—in.	233.7	232.6	231.1
Height—in.	55.6	55.4	54.7
Curb Weight—lbs.	5,250	5,425	5,345
Fuel Capacity—gals.	27	22.5	25
Oil Capacity—qts.	4 (1)	4 (1)	4 (1)
Storage Capacity—cu. ft.	19.27	20.9	20+
Base Price	$9,312	$7,637	$7,062
Price as tested	$11,435	$9,452	$8,737
Engine:	OHV V-8	OHV V-8	OHV V-8
Bore & Stroke—ins.	4.3x4.06	4.36x3.85	4.32x3.75
Displacement—cu. in.	472	460	440
HP @ RPM	205 @ 3600	215 @ 4000	230 @ 4000
Torque: lbs.-ft. @ rpm	365 @ 2000	350 @ 2600	350 @ 3200
Compression Ratio	8.25:1	NA	8.2:1
Carburetion	4V	4V	4V
Transmission	Auto. Turbo Hydra-Matic	Auto. Select Shift	Auto. Torqueflite
Final Drive Ratio	2.93	3.00	3.23 (?)
Steering Type	Recirculating Ball & Nut Power	Recirculating Ball & Nut With Integral Power Unit	Recirculating Ball Power
Steering Ratio	17.8-9.0	21.6 To 1	18.9:1
Turning Diameter (curb-to-curb-ft.)	(Wall To Wall) 24.54′	46.7′	44.69′
Wheel Turns (lock-to-lock)	2.83	3.99	3.5
Tire Size	LR78X15 Steel Belted Radials	LR78X15 Steel Belted Radials	LR78X15 Steel Belted Radial Ply
Brakes	Power Disc/Drum	Power Disc/Drum	Power Disc/Disc
Front Suspension	Coils/Shocks Front Diagonal Tie Struts Stabilizer	Coils/Shocks Axial Strut Stabilizer	Torsion Bar Shocks Stabilizer
Rear Suspension	4 Link, Coils/ Shocks	Three Link, Rubber Cushioned Pivots Coils/Shocks	Leaf Springs Shocks
Body/Frame Construction	Perimeter Frame	Body On Perimeter Frame	Unitized Construction

mtcars data frame

How can you determine what variables are in a data frame?

```
> View(mtcars)        # only works in Rstudio, not in Markdown
> glimpse(mtcars)
> ? mtcars            # this data frame as a code book
```

[, 1]	mpg	Miles/(US) gallon
[, 2]	cyl	Number of cylinders
[, 4]	hp	Gross horsepower
[, 6]	wt	Weight (1000 lbs)
[, 9]	am	Transmission (0 = automatic, 1 = manual)

Do cars that weigh more use more fuel?

Question: do cars that weigh more use more fuel?

What variables in the mtcars data frame are of interest?

- mpg
- wt

We can create a scatter plot using base graphics…

```
> plot(mtcars$wt, mtcars$mpg)
```

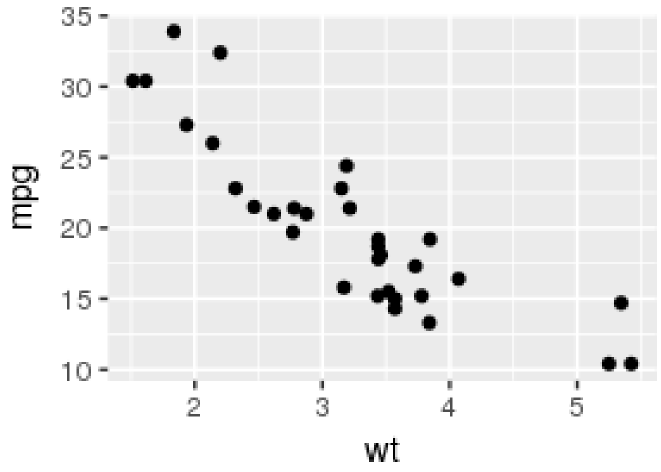
Creating a scatter plot in ggplot

Data frame to be used

Aesthetic mapping

```
> ggplot(data = mtcars, mapping = aes(x = wt, y = mpg)) +
      geom_point()
```

Adds a layer with glyphs



	wt	cyl	hp	mpg	disp
Mazda RX4	2.620	6	110	21.0	160.0
Mazda RX4 Wag	2.875	6	110	21.0	160.0
Datsun 710	2.320	4	93	22.8	108.0
Hornet 4 Drive	3.215	6	110	21.4	258.0
Hornet Sportabout	3.440	8	175	18.7	360.0

Creating a scatter plot in ggplot

Data frame to be used

Aesthetic mapping

```
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```

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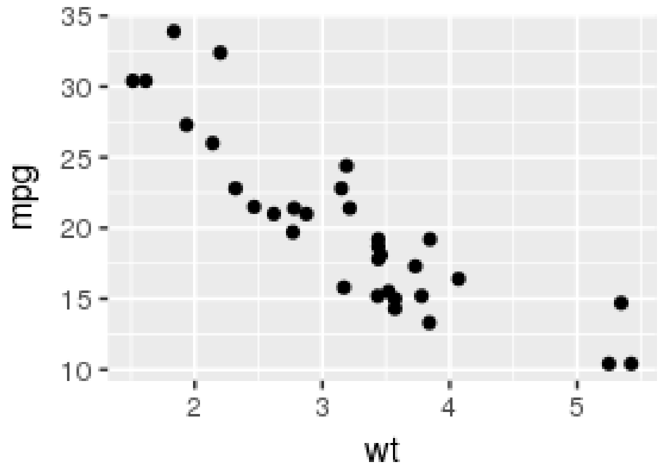

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A lot more that ggplot can do!

- More aesthetic mapping
- Multiple glyphs/layers
- Axis labels
- Facets
- Visual themes
- Different coordinate systems
- Etc.

[The R Graph Gallery]

Let’s try the rest in R!